

Program: Diploma in Integrated Circuit Design and Fabrication	
Course Code: 6498	Course Title: Sensors and Actuators Lab
Semester: 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To be familiar with the use different transducers and sensors
- To make the students aware of the characteristics of transducers.
- To familiarize electrical actuation using Arduino/Raspberry Pi

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principle of resistance, inductance, capacitance, diode, transistor etc.	2039	Fundamentals of Electrical & Electronics Engineering Lab	2
Permeability, Electromagnetic induction, Self and mutual inductance, transformer	3041	Fundamentals of Electrical & and electronics.	3
Develop simple interfacing applications using Arduino.	2049	Electronics Tinkering Workshop	2

Course Outcomes:

On completion of the course, the students will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Plot the characteristics of resistive transducers and measure various parameters using resistive transducers	12	Applying
CO2	Measure various parameters using capacitive, inductive and piezo electric transducers and plot the characteristics of inductive transducers	9	Applying

CO3	Plot the characteristics of photoelectric transducers and measure various parameters using ultra sonic and hall effect transducer	9	Applying
CO4	Develop interfacing circuits using Arduino/Raspberry pi for control of different electrical actuators.	9	Applying
	Lab Exam	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3			
CO2	3	3	3	3			
CO3	3	3	3				
CO4	3	3	2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Plot the characteristics of resistive transducers and measure various parameters using resistive transducers		
M1.01	Measure of displacement using potentiometer.	3	Applying
M1.02	Plot the characteristics of strain gauge.	3	Applying
M1.03	Plot the characteristics of LDR.	3	Applying
M1.04	Plot the characteristics of Thermistor.	3	Applying
CO2	Measure various parameters using capacitive, inductive and piezo electric transducers and plot the characteristics of inductive transducers.		
M2.01	Plot the characteristics of LVDT.	3	Applying
M2.02	Measure level/displacement using capacitive transducer.	3	Applying

M2.03	Measure dynamic pressure/vibration using piezo electric transducer.	3	Applying
	Lab Test – I	3	
CO3	Plot the characteristics of photoelectric transducers and measure various parameters using ultrasonic and hall effect transducer		
M3.01	Plot the characteristics of Photo diode.	3	Applying
M 3.02	Plot the characteristics of Photo transistor.	3	Applying
M3.03	Measure displacement level using ultrasonic transducer.	3	Applying
CO4	Develop interfacing circuits using Arduino/Raspberry pi for control of different electrical actuators.		
M4.01	DC Motor control using Arduino/Raspberry Pi	3	Applying
M4.02	Stepper Motor actuation using Arduino/Raspberry Pi	3	Applying
M4.03	Electromagnetic relay actuation using Arduino/Raspberry Pi	3	Applying
	Lab Exam 2	3	

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 2-3. There is no duplication allowed in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should include the open ended experiment in their lab record.

Text /Reference:

T/R	Book Title/Author
R1	IA.K. Sawhney, “ A course in Electrical and Electronic measurement and Instrumentation”, Dhanpathrayi and Co.
R2	D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd.
R3	B. C. Nakra, K. K. Chaudhry, “Instrumentation, Measurement And Analysis”, Tata McGraw-Hill Publishers. R5 Bela G. Liptak, “Process M
R4	Krishna Kant, “Computer-Based Industrial Control”, PHI.

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=FYggY7LLDrw
2	https://www.slideshare.net/slideshow/classification-of-transducers/39482215
3	https://www.electronics-tutorials.ws/io/io_1.html
4	https://www.ctscorp.com/resource-center/tutorials/piezo-basics/
5	https://www.arduino.cc/
6	https://www.raspberrypi.org/

